A Dual-Orbit Refueling Architecture for Safe, Scalable, and Cost-Effective Human Mars Missions

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Abstract

Human missions to Mars are constrained by launch mass, return-risk uncertainty, and single-mission architectures. This paper proposes a dual-orbit refueling infrastructure consisting of Earth Orbit Fuel Depots and Mars Orbit Fuel Depots, enabling safe crew return, reduced launch mass, and scalable interplanetary operations. By pre-positioning propellant and enabling orbital refueling, Mars missions can transition from high-risk expeditions to repeatable operations.

1. Problem Statement

Current Mars mission concepts face four major limitations: (1) launching fully fueled spacecraft from Earth, (2) limited abort capability, (3) high risk during Mars ascent and Earth return, and (4) one-time mission designs with minimal reusability. Failure in return fuel availability results in mission loss without recovery options.

2. Proposed Architecture

Earth Orbit Fuel Depot: A modular depot in Low or High Earth Orbit enables partial-fuel launch, pre-Mars refueling, and mission flexibility.
Interplanetary Transit: Optimized propulsion with optional gravity assists minimizes propellant usage while maintaining crew safety timelines.
Mars Orbit Fuel Depot: A pre-deployed depot supplied by tanker probes and future ISRU guarantees Mars ascent and Earth return capability.
Earth Return: Refueling in Mars orbit enables Mars-to-Earth transfer without Earth-launched return fuel.

3. Key Advantages

Eliminates single-point failure for Mars return, reduces Earth launch mass and cost, enables reusable Mars infrastructure, improves crew safety, and supports international collaboration.

4. International Contribution Model

ISRO: Mars communication satellites, ISRU precursor missions, orbital fuel storage technology. NASA: Human mission systems, life support, deep-space operations. SpaceX: Heavy-lift launch, reusable interplanetary transport.

5. Conclusion

Pre-positioned orbital propellant transforms Mars exploration into a sustainable transportation system. This architecture enables safe, economical, and repeatable human Mars missions.

Do not send all the fuel with the crew. Send the fuel ahead, store it in orbit, and make Mars missions repeatable.